Nonlinear Convolution Equations, Spectral Rigidity, and the Sharp Restriction Constant on S^1

A Compactness Route to Finiteness of Critical Values

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1 Overview and Strategy

The ultimate target of this programme is the sharp Fourier extension constant on the unit circle $S^1 \subset \mathbb{R}^2$:

$$\mathcal{R} := \sup_{0 \neq f \in L^2(S^1)} \frac{\left\| \widehat{f\sigma} \right\|_{L^6(\mathbb{R}^2)}}{\|f\|_{L^2(S^1)}},$$

where σ is arclength measure on S^1. Every extremiser satisfies a nonlinear convolution equation on the sphere that fits into a general framework studied in the first part of this note.

1.1 The main strategy

After normalising $\|f\|_{L^2} = 1$ and writing $u = \widehat{f\sigma}$, an extremiser satisfies the Euler–Lagrange equation

$$u = c\big((|u|^4 u) * \widehat{\sigma}\big), \qquad c = \Lambda^{-1}, \qquad \Lambda = \|u\|_{L^6}^6 .$$

The strategy is:

1. **Spectral support forcing.** Show that $\widehat{u}$ is supported on S^1, reducing from all of $\mathbb{R}^2$ to a density $\rho \in L^2(S^1)$. **[Proved]**

2. **Fredholm structure.** Prove that the linearisation of the Euler–Lagrange map at any critical point is Fredholm of index zero, so that solutions are locally isolated modulo symmetry. **[Proved]**

3. **Compactness modulo symmetry.** Show that the set of normalised critical points is sequentially compact after quotienting by the symmetry group $G = U(1) \times SO(2) \ltimes \mathbb{R}^2$. **[Open]**

4. **Nondegeneracy.** At each critical point, verify that the kernel of the linearisation equals the tangent space to the symmetry orbit. At the constant solution, this reduces to finitely many Bessel integral inequalities. **[Conditional]**

5. **Finiteness and identification.** Conclude that the critical values Λ_j are finitely many, and identify the sharp constant as $\mathcal{R} = \max_j \Lambda_j^{1/6}$. **[Conditional]**

1.2 Status summary

Component	Status	Key input
Spectral support forcing (any n)	**[Proved]**	Paley–Wiener + support calculus
Compactness of T on PW (general)	**[Proved]**	Rellich for PW spaces
Compactness of K_{u_0} on $L^2(S^1)$	**[Proved]**	Truncation + Hilbert–Schmidt
Fredholm structure ($I - K$ index 0)	**[Proved]**	Riesz–Schauder
1D case (S^0)	**[Proved]**	Explicit algebraic solution
Nondegeneracy at constant solution	**[Conditional]**	Bessel integrals $\mathcal{J}_n$
Existence of maximiser	**[Proved]**	Profile decomposition + strict convexity
Compactness modulo G (all critical pts)	**[Open]**	Ruling out multi-profile splitting
Finiteness of critical values	**[Conditional]**	Items 3 + 4 above
Higher-dimensional generalisations	**[Open]**	Incidence-fibre smoothing

Part I
General Theory

2 Setup and Notation

We work throughout with *entire functions of finite exponential type*. An entire function $u : \mathbb{C} \to \mathbb{C}$ has *exponential type* $\tau \geq 0$ if $|u(z)| \leq Ce^{\tau|z|}$ for some $C > 0$ and all $z \in \mathbb{C}$. The *Paley–Wiener–Schwartz theorem* identifies this class: u is entire of type $\leq \tau$ with $u|_{\mathbb{R}} \in L^2(\mathbb{R})$ if and only if $\widehat{u} \in L^2$ is supported in $[-\tau, \tau]$. We write PW_τ for the corresponding Hilbert space.

Fourier conventions:

$$\widehat{u}(\xi) = \int_{\mathbb{R}} u(x)e^{-ix\xi}\, dx, \qquad u(x) = \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{u}(\xi)e^{ix\xi}\, d\xi,$$

so $\widehat{u * v} = \widehat{u} \cdot \widehat{v}$ and $\widehat{u \cdot v} = \frac{1}{2\pi}\widehat{u} * \widehat{v}$.

Throughout, $k \geq 1$ is a fixed positive integer, $c \in \mathbb{C} \setminus \{0\}$ is a fixed constant, and f is an entire function of exponential type $\tau_f > 0$ whose restriction to $\mathbb{R}$ lies in $L^2(\mathbb{R})$.

3 The Two Convolution Equations

3.1 The equation $u^k = c\,(u * f)$: spectral expansion kills solutions

In Fourier space the equation reads $\widehat{u}^{*k}(\xi) = c\,\widehat{u}(\xi)\widehat{f}(\xi)$.

Proposition 3.1 (Type obstruction)**.** *Let $u \in \mathrm{PW}_\tau$ with $\tau > 0$ satisfy $u^k = c(u * f)$ for $k \geq 2$. Then $u \equiv 0$.*

Proof. The Titchmarsh convolution theorem gives $\mathrm{supp}(\widehat{u}^{*k}) \subseteq [-k\tau, k\tau]$, while the right side is supported in $[-\tau, \tau]$. For $k \geq 2$ this forces $\tau = 0$, so u is a polynomial; degree comparison then forces u to be constant, and the only constants satisfying the equation are the $(k-1)$-th roots of $c\widehat{f}(0)$ together with 0. □

Corollary 3.2. *For $k \geq 2$ the solution set of $u^k = c(u * f)$ in the Paley–Wiener class is a finite discrete set of at most k constants.*

3.2 The equation $u = c\,(u^k * f)$: compact fixed-point structure

We now turn to the equation

$$u = c\,(u^k * f). \tag{1}$$

In Fourier space:

$$\widehat{u}(\xi) = c\,\widehat{u}^{*k}(\xi) \cdot \widehat{f}(\xi). \tag{2}$$

Proposition 3.3 (Type self-consistency)**.** *Any solution $u \in \mathrm{PW}_\tau$ of (1) satisfies $\tau \leq \tau_f$.*

Proof. The right side of (2) is supported in $\mathrm{supp}(\widehat{u}^{*k}) \cap \mathrm{supp}(\widehat{f}) \subseteq [-\tau_f, \tau_f]$. □

This is the key contrast: the equation $u = c(u^k * f)$ does *not* force $\tau = 0$; solutions of positive exponential type exist as long as $\tau \leq \tau_f$.

Proposition 3.4 (Compactness of T on Paley–Wiener spaces)**.** *Define $T : \mathrm{PW}_{\tau_f} \to \mathrm{PW}_{\tau_f}$ by $T(v) = c(v^k * f)$. Then T maps bounded sets in PW_τ (for any $\tau > \tau_f$) to precompact sets in PW_τ. In particular, $T : \mathrm{PW}_\tau \to \mathrm{PW}_\tau$ is a compact operator.*

Proof. Since $v \in \mathrm{PW}_{\tau_f}$ implies $v^k \in \mathrm{PW}_{k\tau_f}$, the function $v^k * f$ has Fourier support in $\mathrm{supp}(\widehat{v^k}) \cap \mathrm{supp}(\widehat{f}) \subseteq [-\tau_f, \tau_f]$. Hence T maps PW_τ into PW_{τ_f}, and the inclusion $\mathrm{PW}_{\tau_f} \hookrightarrow \mathrm{PW}_\tau$ is compact by the Rellich-type theorem for Paley–Wiener spaces. □

3.3 Fredholm structure and local manifold theorem

At a solution u_0 of (1), the Fréchet derivative of $G(u) := u - c(u^k * f)$ is

$$DG(u_0)[h] = h - ck(u_0^{k-1}h * f) =: (I - K)[h],$$

where $K : h \mapsto ck(u_0^{k-1}h * f)$ is compact by the same argument. By Riesz–Schauder, $I - K$ is **Fredholm of index zero**.

Theorem 3.5 (Local manifold structure)**.** *Let u_0 be a solution of* (1) *at which $DG(u_0) = I - K$ is invertible. Then u_0 is an isolated solution. More generally, the solution set near u_0 is a smooth manifold of dimension* $\dim\ker(I - K)$*, which is finite.*

Remark 3.6 (Direct contraction argument). For u, v in a ball of radius R in PW_{τ_f}, Young's inequality gives $\|T(u) - T(v)\| \leq |c|\, kR^{k-1} \|f\|_{L^1} \|u - v\|$. This is a contraction whenever $R < (|c|\, k \|f\|_{L^1})^{-1/(k-1)}$, giving local uniqueness without spectral theory.

3.4 Comparison of the two equations

	$u^k = c(u * f)$	$u = c(u^k * f)$
Fourier identity	$\widehat{u}^{*k} = c\widehat{u}\widehat{f}$	$\widehat{u} = c\widehat{u}^{*k} \cdot \widehat{f}$
Type constraint ($k \geq 2$)	Forces $\tau = 0$ (Titchmarsh)	Allows $\tau \leq \tau_f$
Operator structure	No compactness	T compact
Linearisation	Not Fredholm in general	$I - K$, Fredholm index 0
Solutions ($k \geq 2$, generic)	Finitely many constants	Isolated; Leray–Schauder applies

Table 1: Analytic contrast between the two convolution equations.

4 Spectral Support Forcing

We specialise to $f = \widehat{\sigma}$, where σ is the surface measure on $S^{n-1} \subset \mathbb{R}^n$. By the Paley–Wiener theorem, f is entire of exponential type 1, and $\widehat{f} = (2\pi)^n\sigma$ as a distribution.

Proposition 4.1 (Spectral support forcing)**.** *If u is entire of exponential type satisfying $u = c(u^k * \widehat{\sigma})$, then $\widehat{u}$ is a distribution supported on S^{n-1}. The same conclusion holds for the equation $u = c(|u|^{2(k-1)}u * \widehat{\sigma})$.*

Proof. From (2), $\widehat{u} = c\,\widehat{u}^{*k} \cdot (2\pi)^n \sigma$. The right side is supported on $\operatorname{supp}\sigma = S^{n-1}$.

For the $|u|^{2(k-1)}u$ variant: writing $|u|^{2(k-1)}u = u^k \bar{u}^{k-1}$ in Fourier space involves $(k-1)$-fold convolutions of $\widehat{u}$, where $\widehat{\bar{u}}(\xi) = \overline{\widehat{u}(-\xi)}$. Since S^{n-1} is symmetric under $\xi \mapsto -\xi$, the support of all convolved terms lies in the sumset of copies of $\pm S^{n-1}$, and multiplication by σ restricts back to S^{n-1}. □

Remark 4.2. In dimension $n \geq 2$, the conclusion $\operatorname{supp}\widehat{u} \subseteq S^{n-1}$ means u is a *Herglotz wave function*, satisfying $(-\Delta - 1)u = 0$.

4.1 Reduction to a density equation on the sphere

Write $\widehat{u} = \rho\, d\sigma$ for $\rho \in L^2(S^{n-1})$, so

$$u(x) = \int_{S^{n-1}} \rho(\omega) e^{ix\cdot\omega}\, d\sigma(\omega).$$

The equation $u = c(u^k * \widehat{\sigma})$ then reduces to the *spherical fixed-point equation*

$$\rho(\omega) = c \int_{V_k(\omega)} \rho(\omega_1) \cdots \rho(\omega_k)\, d\sigma_{\text{fiber}}(\omega_1, \ldots, \omega_k), \qquad \sigma\text{-a.e. } \omega \in S^{n-1}, \tag{3}$$

where $V_k(\omega) = \{(\omega_1, \ldots, \omega_k) \in (S^{n-1})^k : \omega_1 + \cdots + \omega_k = \omega\}$ is the incidence fibre.

Warning 4.3 (Incidence fibre $\neq$ spherical convolution). The operator $T(\rho)(\omega) = \int_{V_k(\omega)} \rho^{\otimes k}\, d\sigma_{\text{fiber}}$ is *not* the same as convolution by the spherical measure. In particular, the Funk–Hecke theorem does not diagonalise T in spherical harmonics. Any reduction to finitely many harmonic coefficients requires separate justification.

5 The One-Dimensional Case: Complete Solution

For $n = 1$, $S^0 = \{-1, +1\}$, $\sigma = \delta_{-1} + \delta_{+1}$, and $f(x) = \widehat{\sigma}(x) = 2\cos x$. Proposition 4.1 forces $\widehat{u} = a\delta_{-1} + b\delta_{+1}$, so $u(x) = ae^{-ix} + be^{ix}$.

5.1 Algebraic reduction

Expanding u^k and computing $u^k * f$ picks out frequencies ± 1, which exist only when k is **odd**.

Proposition 5.1 (Parity obstruction)**.** *If k is even, the equation $u = c(u^k * f)$ with $f = 2\cos$ has only $u \equiv 0$.*

For k odd, setting $\beta_k := \binom{k}{(k-1)/2}$ and comparing coefficients of $e^{\mp ix}$ gives the single constraint

$$ab = \lambda_k, \qquad \lambda_k := \left(\frac{1}{2\pi c \beta_k}\right)^{2/(k-1)}. \tag{4}$$

Theorem 5.2 (Complete solution in 1D)**.** ***[Proved]*** *Let $k \geq 3$ be odd, $c \in \mathbb{C} \setminus \{0\}$, $f = 2\cos$. The solution set of $u = c(u^k * f)$ in the Paley–Wiener class is*

$$\{0\} \cup \bigsqcup_{j=0}^{(k-3)/2} \{(a,b) \in \mathbb{C}^2 : ab = \omega^j \lambda_k\}, \qquad \omega = e^{4\pi i/(k-1)}.$$

This is a finite union of $(k-1)/2$ smooth one-dimensional complex manifolds (each $\cong \mathbb{C}^*$), plus the origin. Each component is exactly the translation orbit of any single non-trivial solution on it: the translation $u(x) \mapsto u(x-t)$ acts by $(a,b) \mapsto (e^{-it}a, e^{it}b)$, preserving ab.

Example 5.3 ($k = 3$)**.** $\beta_3 = 3$. Constraint: $ab = (6\pi c)^{-1}$. One component, the translation orbit of $u(x) = (6\pi c)^{-1/2} \cos x$.

Example 5.4 ($k = 5$)**.** $\beta_5 = 10$. Constraint: $ab = (20\pi c)^{-1/2}$. Two components (the scaling group $\mu_2 = \{1, -1\}$ generates a second hyperbola).

Remark 5.5 (Real solutions)*.* For real-valued solutions $b = \bar{a}$, so $ab = |a|^2 \geq 0$. Only the hyperbola with $\lambda_k \in \mathbb{R}_{>0}$ contributes; the real solution manifold is the circle $|a| = |\lambda_k|^{1/2}$, parametrised by translations.

Part II
The S^1 Restriction Problem

6 The Sharp Restriction Constant and Its Euler–Lagrange Equation

6.1 Setup

The sharp Fourier extension constant on $S^1 \subset \mathbb{R}^2$ is

$$\mathcal{R} := \sup_{\substack{f \in L^2(S^1) \\ f \neq 0}} \frac{\|Ef\|_{L^6(\mathbb{R}^2)}}{\|f\|_{L^2(S^1)}}, \qquad (Ef)(x) := \int_{S^1} f(\omega) e^{ix\cdot\omega}\, d\sigma(\omega),$$

where σ is the arclength measure on S^1 (total mass 2π). Every extension $u = Ef$ satisfies $(-\Delta - 1)u = 0$.

6.2 The Euler–Lagrange equation

Normalise $\|f\|_{L^2(S^1)} = 1$. A critical point of $f \mapsto \|Ef\|_{L^6}^6$ on the unit sphere satisfies

$$E^*(|u|^4 u) = \Lambda f, \qquad u = Ef, \qquad \Lambda = \|u\|_{L^6}^6\,, \tag{5}$$

where $E^* : L^{6/5}(\mathbb{R}^2) \to L^2(S^1)$ is the adjoint restriction: $(E^*g)(\omega) = \int_{\mathbb{R}^2} g(x)e^{-ix\cdot\omega}\,dx$ restricted to $\omega \in S^1$. Equivalently,

$$u = c\big((|u|^4 u) * \widehat{\sigma}\big), \qquad c = \Lambda^{-1}. \tag{6}$$

Warning 6.1 (The nonlinearity is $|u|^4 u$, not u^5). The restriction problem gives $|u|^4 u = u^3\bar{u}^2$, which is degree 5 but *not* holomorphic. The linearisation is *real-linear*, not complex-linear: the Jacobian couples the Fourier mode n to mode $-n$. This distinction is essential for the spectral analysis in Section 8.

6.3 The symmetry group

The natural symmetry group is

$$G = U(1) \times SO(2) \ltimes \mathbb{R}^2, \qquad \dim G = 4.$$

It acts on densities by

$$(e^{i\alpha}, R, x_0) \cdot f(\omega) = e^{i\alpha} e^{-ix_0\cdot\omega} f(R^{-1}\omega).$$

Phase, rotation, and modulation/translation all preserve $\|f\|_{L^2}$ and $\|Ef\|_{L^6}$, hence preserve Λ. Every critical point generates a G-orbit of critical points with the same Λ.

Definition 6.2 (Shape)**.** A *shape* is an equivalence class of normalised critical points modulo the G-action. Two critical points with the same shape have the same critical value Λ.

6.4 What "finding all critical values" means

The search problem is: determine the finite set

$$\mathcal{Z} := \{(f, \Lambda) : E^*(|Ef|^4 Ef) = \Lambda f,\ \|f\|_2 = 1\}/G.$$

If $\mathcal{Z}$ is finite, the sharp restriction constant is

$$\mathcal{R} = \max_{[f,\Lambda]\in\mathcal{Z}} \Lambda^{1/6} = c_{\min}^{-1/6},$$

where $c_{\min}$ is the smallest positive $c_j = \Lambda_j^{-1}$. The normalisation $\|f\|_2 = 1$ is essential: without it, αu solves the equation with c replaced by $c\alpha^{-4}$, preventing discreteness.

7 Fredholm Structure of the Linearisation

7.1 The extension operator E is not compact

One might hope that $E : L^2(S^1) \to L^6(\mathbb{R}^2)$ is compact, which would immediately give compactness of the nonlinear map $T(f) = E^*(|Ef|^4 Ef)$. This is **false**.

Proposition 7.1 (E is not compact)**.** ***[Proved]*** *The extension operator $E : L^2(S^1) \to L^6(\mathbb{R}^2)$ is bounded (Stein–Tomas) but not compact.*

Proof. Fix $f \in L^2(S^1)$ with $\|f\|_2 = 1$ and $\|Ef\|_{L^6} > 0$. For $n \in \mathbb{N}$, define $f_n(\omega) := e^{-ine_1\cdot\omega} f(\omega)$ where $e_1 = (1, 0)$. Then $\|f_n\|_2 = 1$ and $Ef_n(x) = Ef(x-ne_1) = u(x-ne_1)$, a translate of $u = Ef$.

For $n \neq m$ with $|n - m|$ large, the supports of $|u(\cdot - ne_1)|$ and $|u(\cdot - me_1)|$ in any L^6-significant region are essentially disjoint (since $u(x) \to 0$ as $|x| \to \infty$ by Riemann–Lebesgue), so

$$\|Ef_n - Ef_m\|_{L^6} = \|u(\cdot - ne_1) - u(\cdot - me_1)\|_{L^6} \xrightarrow{|n-m|\to\infty} 2^{1/6} \|u\|_{L^6} > 0.$$

Hence $\{Ef_n\}$ has no Cauchy subsequence. □

The obstruction is the noncompact translation symmetry: high-frequency modulations on S^1 shift the extension arbitrarily far in x-space. This means compactness of the critical set requires quotienting by G; it cannot be deduced from a single operator bound.

7.2 The linearised operator K_{u_0} is compact

Despite the non-compactness of E, the *specific* operator appearing in the linearisation is compact, thanks to the decay of the solution u_0.

Definition 7.2. At a critical point f_0 with $u_0 = Ef_0$, define the linearised operator $K = K_{u_0} : L^2(S^1) \to L^2(S^1)$ by

$$Kh := E^*\big(DN(u_0)[Eh]\big),$$

where $DN(u_0)[v] = 3|u_0|^4 v + 2|u_0|^2 u_0^2 \bar{v}$ is the (real-linear) Fréchet derivative of $N(u) = |u|^4 u$.

Theorem 7.3 (Compactness of K_{u_0})**.** ***[Proved]*** *For any $u_0 = Ef_0$ with $f_0 \in L^2(S^1)$, the operator $K_{u_0} : L^2(S^1) \to L^2(S^1)$ is compact.*

Proof. We prove compactness for the "main term" $K_1 h := E^*(|u_0|^4 Eh)$; the argument for $K_2 h := E^*(|u_0|^2 u_0^2 \overline{Eh})$ is identical, and $K = 3K_1 + 2K_2$.

Step 1: Truncation. For $R > 0$ let χ_R denote the indicator of $B(0, R) \subset \mathbb{R}^2$ and set $K_1^R h := E^*(\chi_R\, |u_0|^4\, Eh)$. Then

$$\big\|K_1 - K_1^R\big\|_{L^2\to L^2} \le \|E^*\|_{L^{6/5}\to L^2}\ \big\|(1 - \chi_R)|u_0|^4\big\|_{L^{3/2}}\ \|E\|_{L^2\to L^6}\,.$$

Here $\|E\|$ and $\|E^*\|$ are the (finite) Stein–Tomas constants, and

$$\big\|(1 - \chi_R)|u_0|^4\big\|_{L^{3/2}}^{3/2} = \int_{|x|>R} |u_0|^6\, dx \xrightarrow{R\to\infty} 0,$$

since $u_0 \in L^6(\mathbb{R}^2)$. Hence $\left\|K_1 - K_1^R\right\|_{\mathrm{op}} \to 0$.

Step 2: K_1^R is Hilbert–Schmidt. The integral kernel of K_1^R on $S^1 \times S^1$ is

$$\mathcal{K}_R(\omega, \omega') = \int_{B(0,R)} |u_0(x)|^4 \, e^{ix\cdot(\omega'-\omega)} \, dx.$$

Since $|u_0|^4 \chi_R \in L^1(\mathbb{R}^2) \cap L^\infty(\mathbb{R}^2)$ (bounded and compactly supported), its Fourier transform is continuous on $\mathbb{R}^2$. Therefore $\mathcal{K}_R$ is continuous on the compact set $S^1 \times S^1$, hence bounded, and in particular belongs to $L^2(S^1 \times S^1)$. Thus K_1^R is Hilbert–Schmidt, hence compact.

Step 3: Conclusion. $K_1 = \lim_{R\to\infty} K_1^R$ in operator norm, where each K_1^R is compact. Hence K_1 is compact. □

Corollary 7.4 (Fredholm structure)**.** ***[Proved]*** *At any critical point f_0 of the restriction functional with $\|f_0\|_2 = 1$ and $\Lambda_0 = \|Ef_0\|_6^6 > 0$, the operator*

$$L := K_{u_0} - \Lambda_0 I$$

has the form $L = K - \Lambda_0 I$ with K compact. Therefore $L : L^2(S^1) \to L^2(S^1)$ is Fredholm of index zero. In particular, $\ker L$ is finite-dimensional, and L is invertible if and only if $\ker L = \{0\}$.

Remark 7.5 (Self-adjointness). With respect to the real inner product $\operatorname{Re}\langle f, g\rangle_{L^2(S^1)}$, the operator K is self-adjoint. Indeed, $DN(u_0)$ is self-adjoint on $L^2(\mathbb{R}^2, \operatorname{Re}\langle\cdot,\cdot\rangle)$ (it is the Hessian of the real functional $u \mapsto \frac{1}{6}|u|^6$), and E^* is the adjoint of E. Consequently, the eigenvalues of L are real.

8 Nondegeneracy at the Constant Solution

8.1 Setup

The constant function $f_0 = (2\pi)^{-1/2}$ has $\|f_0\|_{L^2(S^1)} = 1$. Its extension is the radial function

$$u_0(x) = Ef_0(x) = (2\pi)^{1/2} J_0(|x|),$$

where J_0 is the Bessel function of order zero. Set $A := (2\pi)^{1/2}$, so $u_0(r) = AJ_0(r)$. The critical value is

$$\Lambda = \|u_0\|_{L^6}^6 = A^6 \cdot 2\pi \int_0^\infty J_0(r)^6 \, r \, dr = 16\pi^4 \int_0^\infty J_0(r)^6 \, r \, dr.$$

The stabiliser of f_0 under G is $\mathrm{Stab}_G(f_0) = SO(2)$ (rotations of S^1 fix the constant function). The orbit has dimension $\dim G - \dim \mathrm{Stab} = 4 - 1 = 3$, generated by phase ($U(1)$) and translations ($\mathbb{R}^2$).

8.2 Fourier decomposition of the linearised operator

Since u_0 is radial, the linearised operator $K : L^2(S^1) \to L^2(S^1)$ preserves each pair of Fourier modes $\{e^{in\theta}, e^{-in\theta}\}$. We use the Fourier expansion

$$h(\theta) = \sum_{m\in\mathbb{Z}} c_m e^{im\theta}, \qquad Eh(r,\phi) = \sum_{m\in\mathbb{Z}} 2\pi\, i^{|m|} c_m\, J_{|m|}(r)\, e^{im\phi},$$

where (r, ϕ) are polar coordinates in $\mathbb{R}^2$.

Proposition 8.1 (Eigenvalue structure)**.** ***[Proved]*** *Define the Bessel integrals*

$$\mathcal{J}_n := \int_0^\infty J_0(r)^4\, J_n(r)^2\, r\, dr, \qquad n \geq 0, \tag{7}$$

and $\mathcal{I}_n := 16\pi^4 \mathcal{J}_n$, so that $\Lambda = \mathcal{I}_0$. Then, on the sector $\{c_n e^{in\theta} + c_{-n} e^{-in\theta}\}$ with $n \geq 1$, the operator $K = E^* DN(u_0)E$ acts as a 4×4 real-linear map with eigenvalues

$$5\mathcal{I}_n \quad (\textit{multiplicity } 2) \qquad \textit{and} \qquad \mathcal{I}_n \quad (\textit{multiplicity } 2).$$

On the sector $n = 0$, the eigenvalues are $5\mathcal{I}_0$ (for real perturbations) and $\mathcal{I}_0$ (for imaginary perturbations), each with multiplicity 1.

Proof. Write $v = Eh$ where $h = c_n e^{in\theta} + c_{-n} e^{-in\theta}$. Then $v = 2\pi i^n J_n(r)[c_n e^{in\phi} + c_{-n} e^{-in\phi}]$ (using $E(e^{im\theta}) = 2\pi i^{|m|} J_{|m|}(r) e^{im\phi}$).

Since $u_0 = AJ_0(r)$ is real and radial, $DN(u_0)[v] = 3u_0^4 v + 2u_0^4 \bar{v}$. The $e^{in\phi}$ component of $DN(u_0)[v]$ is $2\pi A^4 J_0^4 J_n[3i^n c_n + 2(-i)^n \bar{c}_{-n}]$, and the $e^{-in\phi}$ component is $2\pi A^4 J_0^4 J_n[3i^n c_{-n} + 2(-i)^n \bar{c}_n]$.

Applying E^* and using the identity $\langle E^* g, e^{im\theta}\rangle = 2\pi(-i)^{|m|} \int_0^\infty g_m(r) J_{|m|}(r)\, r\, dr$ (where g_m is the m-th angular Fourier coefficient of g), together with $(-i)^n i^n = 1$ and $(-i)^{2n} = (-1)^n$, one computes:

$$(Kh)_n = \mathcal{I}_n \big[3c_n + 2(-1)^n \bar{c}_{-n}\big], \tag{8}$$

$$(Kh)_{-n} = \mathcal{I}_n \big[3c_{-n} + 2(-1)^n \bar{c}_n\big]. \tag{9}$$

Writing $c_n = p + iq$, $c_{-n} = s + it$ (real and imaginary parts), the system decouples into a *real part* and an *imaginary part*, each a 2×2 real system. For both parities of n, both 2×2 matrices have the same eigenvalues: 5 and 1 (with the eigenvectors exchanging between the real and imaginary blocks as $(-1)^n$ changes sign). Hence the eigenvalues of K on the sector are $5\mathcal{I}_n$ and $\mathcal{I}_n$, each with multiplicity 2. The $n = 0$ case is analogous with multiplicities halved. □

8.3 Identification of the symmetry kernel

Proposition 8.2 (Symmetry directions)**.** ***[Proved]*** *The kernel of $L = K - \Lambda I$ on $L^2(S^1)$ includes the following directions, with the indicated eigenvalue identities.*

1. ***Phase*** *($n = 0$, imaginary): $h = if_0$. This lies in the $\mathcal{I}_0$-eigenspace of K. Since $\Lambda = \mathcal{I}_0$, the L-eigenvalue is 0.*

2. ***Translations*** *($n = 1$): $h = -i(\hat{x}_0 \cdot \omega)f_0$ for $\hat{x}_0 \in \mathbb{R}^2$. This lies in the $5\mathcal{I}_1$-eigenspace of K. Since translations are an exact symmetry, $5\mathcal{I}_1 = \Lambda$.*

3. ***No rotation direction****: $\partial_\theta f_0 = 0$ since f_0 is constant.*

In particular, $\mathcal{I}_1 = \Lambda/5$, i.e.,

$$5\int_0^\infty J_0(r)^4\, J_1(r)^2\, r\, dr = \int_0^\infty J_0(r)^6\, r\, dr. \tag{10}$$

This is a Bessel function identity forced by translation invariance.

Proof. (i) is verified by direct computation: $E(if_0) = iu_0$, $DN(u_0)[iu_0] = 3u_0^4(iu_0) + 2u_0^4(-iu_0) = iu_0^5$, and $E^*(iu_0^5) = iE^*(u_0^5) = i\Lambda f_0 = \Lambda(if_0)$.

(ii) The translation $f \mapsto e^{-ix_0\cdot\omega}f$ gives $h = -i(\hat{x}_0 \cdot \omega)f_0$ at first order. The corresponding extension perturbation is $Eh = -\hat{x}_0 \cdot \nabla u_0$ (verified by differentiating $E(e^{-ix_0\cdot\omega}f_0)(x) = u_0(x - x_0)$ in x_0). Differentiating the Euler–Lagrange equation $E^*(|u_0(\cdot - x_0)|^4 u_0(\cdot - x_0)) = \Lambda e^{-ix_0\cdot\omega}f_0$ in x_0 at $x_0 = 0$ gives $Kh = \Lambda h$, confirming that h is in $\ker L$.

From the eigenvalue structure, $h = -i(\hat{x}_0 \cdot \omega)f_0$ has Fourier support on $n = \pm 1$ with $c_{-1} = -\bar{c}_1$. Substituting into (8)–(9) shows this lies in the $5\mathcal{I}_1$-eigenspace, whence $5\mathcal{I}_1 = \Lambda$.

(iii) f_0 is constant, so $\partial_\theta f_0 = 0$. □ □

Remark 8.3 (Euler identity). The identity $Kf_0 = 5\Lambda f_0$ (in the $n = 0$ real sector) follows from the degree-5 homogeneity of N: $DN(u)[u] = 5N(u)$ for all u. Hence $Kf_0 = E^*(DN(u_0)[u_0]) = 5E^*(N(u_0)) = 5\Lambda f_0$.

8.4 The nondegeneracy criterion

The eigenvalues of $L = K - \Lambda I$ on the n-th sector are $5\mathcal{I}_n - \Lambda$ and $\mathcal{I}_n - \Lambda$. At $n = 0$: 0 (phase) and 4Λ (amplitude). At $n = 1$: 0 (translations, multiplicity 2) and $-4\Lambda/5$ (multiplicity 2).

Theorem 8.4 (Nondegeneracy reduces to Bessel integrals)**.** ***[Conditional]*** *The constant solution $f_0 = (2\pi)^{-1/2}$ is nondegenerate modulo G (i.e., $\ker L = T_{f_0}(G \cdot f_0)$ on the tangent space to $\|f\|_2 = 1$) if and only if*

$$\boxed{\mathcal{I}_n \neq \Lambda \quad \text{and} \quad 5\mathcal{I}_n \neq \Lambda \qquad \text{for all } n \geq 2.} \tag{11}$$

Equivalently, in terms of the Bessel integrals (7)*:*

$$\mathcal{J}_n \neq \mathcal{J}_0 \quad \text{and} \quad 5\mathcal{J}_n \neq \mathcal{J}_0 \qquad \text{for all } n \geq 2.$$

Proof. By Proposition 8.1, the eigenvalues of L on the n-th sector ($n \geq 2$) are $5\mathcal{I}_n - \Lambda$ and $\mathcal{I}_n - \Lambda$, each with multiplicity 2. These are both nonzero if and only if (11) holds. The symmetry directions exhaust the kernel at $n = 0$ and $n = 1$ by Proposition 8.2. □

8.5 Asymptotic analysis of $\mathcal{J}_n$

Lemma 8.5 (Decay of $\mathcal{J}_n$)**.** ***[Proved]*** *$\mathcal{J}_n \to 0$ as $n \to \infty$. More precisely, $\mathcal{J}_n = O(n^{-1})$.*

Proof. For $r < n(1-\varepsilon)$, the Bessel function $J_n(r)$ is exponentially small. For $r \geq n$, the uniform bound $|J_0(r)| \leq 1$ and the asymptotic $|J_\nu(r)| \leq Cr^{-1/2}$ (valid for $r \geq \nu$ and all $\nu \geq 0$) give

$$\int_n^\infty J_0(r)^4\, J_n(r)^2\, r\, dr \leq C \int_n^\infty r^{-2} \cdot r^{-1} \cdot r\, dr = C \int_n^\infty r^{-2}\, dr = C/n. \qquad \square$$

Since $\mathcal{J}_n \to 0$ while $\mathcal{J}_0 > 0$ and $\mathcal{J}_0/5 > 0$, the conditions (11) hold for all sufficiently large n. Only finitely many values of n need to be checked.

Lemma 8.6 (Parity structure of the asymptotics)**.** *For moderate n (before the J_n-support retreats past the main lobe of J_0), the large-r average of $J_0(r)^4 J_n(r)^2$ depends on the parity of n. Using the asymptotics $J_\nu(r) \sim \sqrt{2/\pi r}\cos(r - \nu\pi/2 - \pi/4)$ for $r \gg \nu$:*

- ***Even** n:* $\cos(r - n\pi/2 - \pi/4) = \pm\cos(r - \pi/4)$*, so* $\langle J_0^4 J_n^2\rangle_{\text{avg}} \sim (2/\pi r)^3 \cdot \frac{5}{16}$*, the same coefficient as* $\langle J_0^6\rangle$*.*
- ***Odd** n:* $\cos(r - n\pi/2 - \pi/4) = \pm\sin(r - \pi/4)$*, so* $\langle J_0^4 J_n^2\rangle_{\text{avg}} \sim (2/\pi r)^3 \cdot \frac{1}{16}$*, which is $\frac{1}{5}$ of the even case.*

Consequently, for moderate even n, $\mathcal{J}_n$ is close to $\mathcal{J}_0$ (deficiency only from the small-r region where $J_n(r) \ll J_0(r)$), while for moderate odd n, $\mathcal{J}_n$ is close to $\mathcal{J}_0/5 = \mathcal{J}_1$.

Remark 8.7 (Which conditions are tight?). The identity $\mathcal{J}_1 = \mathcal{J}_0/5$ (from translation invariance) shows that the condition $5\mathcal{J}_n = \mathcal{J}_0$ *is* achieved at $n = 1$. For even n, the condition $\mathcal{J}_n = \mathcal{J}_0$ is never achieved ($\mathcal{J}_n < \mathcal{J}_0$ due to the small-r deficiency), but the margin may be small for $n = 2$. The most delicate check is therefore whether $5\mathcal{J}_n = \mathcal{J}_0$ for some *even* $n \geq 2$ (where $\mathcal{J}_n$ is close to $\mathcal{J}_0$, hence $5\mathcal{J}_n$ is close to $5\mathcal{J}_0 \gg \Lambda$, making this condition safe), and whether $\mathcal{J}_n = \mathcal{J}_0$ for some *odd* $n \geq 3$ (where $\mathcal{J}_n \approx \mathcal{J}_0/5$, well below $\mathcal{J}_0$, also safe).

Upshot: the asymptotic structure makes the nondegeneracy conditions (11) very plausible, but does not close the argument. A rigorous verification requires evaluating $\mathcal{J}_n$ for $n = 2, 3, \ldots, n_*$ (where n_* is the threshold beyond which the $O(1/n)$ bound suffices) and confirming the two inequalities at each. This is a finite numerical computation that can be made rigorous with interval arithmetic.

8.6 Nondegeneracy at non-constant solutions

If there exist non-constant critical points of the restriction functional, their nondegeneracy must be established separately. At a non-constant f_0, the extension $u_0 = Ef_0$ is not radial, so K_{u_0} does not decompose into Fourier sectors. Two approaches are available:

Direct spectral analysis. If u_0 has a discrete symmetry (e.g., if f_0 is a real-valued even function on S^1), the problem decomposes into finitely many symmetry sectors, each finite-dimensional.

Generic transversality (Sard–Smale). Introduce a family of G-equivariant perturbations $F_p(\rho) = \rho - cT_p(\rho)$ and show that the parameter derivatives span the cokernel of $DF_p(\rho)$ at every zero. This gives nondegeneracy for a generic parameter p. To cover the original (unperturbed) problem, one must either verify that the spherical measure is a regular parameter, or continue the generic solution set to the original parameter without bifurcation.

Remark 8.8 (Varying c alone is insufficient). As noted in Warning 4.3, varying only the scalar c provides a one-parameter family, which is generically too small for Sard–Smale (the cokernel at a degenerate zero may have dimension > 1). A richer family of perturbations is needed.

9 Compactness Modulo Symmetry

9.1 Regularity bootstrap

Proposition 9.1 (Smoothness and uniform bounds for critical points)**.** ***[Proved]*** *If $f \in L^2(S^1)$ satisfies $E^*(|Ef|^4 Ef) = \Lambda f$ with $\Lambda > 0$, then $f \in C^\infty(S^1)$. Moreover, for each $k \geq 0$ there exists $C_k = C_k(\varepsilon)$ such that $\|f\|_{C^k(S^1)} \leq C_k$ for all normalised critical points with $\Lambda \geq \varepsilon$.*

Proof sketch. Write $u = Ef$, $g = |u|^4 u$. Since $\widehat{u} = f\, d\sigma$ is supported on S^1, the Fourier transform $\widehat{g}$ is supported in the Minkowski sum $S^1 + S^1 + S^1 + S^1 + S^1 = \overline{B(0,5)}$ (using $|u|^4 u = u^3 \bar{u}^2$ and $\widehat{\bar{u}}$ supported on $-S^1 = S^1$). Thus g is *frequency-localised* to $B(0,5)$.

The Euler–Lagrange equation gives $f = \Lambda^{-1} \widehat{g}\big|_{S^1}$, so f inherits the regularity of $\widehat{g}$ restricted to the compact curve $S^1 \subset B(0,5)$. The n-th Fourier coefficient of f on S^1 is

$$\widehat{f}(n) = \Lambda^{-1} \cdot 2\pi(-i)^{|n|} \int_0^\infty g_n(r)\, J_{|n|}(r)\, r\, dr, \tag{12}$$

where $g_n(r)$ is the n-th angular Fourier coefficient of g. For $|n| \gg 1$, $J_{|n|}(r)$ is exponentially small for $r < |n|/2$ (Debye bounds), and the contribution from $r > |n|/2$ is controlled by the $L^{6/5}$ tail of g via Hölder and the Bessel asymptotics $J_\nu(r) = O(r^{-1/2})$ for $r > \nu$. An inductive argument (improved decay of $\widehat{f}(n)$ feeds back into improved regularity of f and hence of u and g) yields $|\widehat{f}(n)| = O(|n|^{-N})$ for all N. All bounds depend only on ε and C_{ST}. □

Corollary 9.2 (Uniform pointwise decay)**.** *For all normalised critical points with $\Lambda \geq \varepsilon$,*

$$|Ef(x)| \leq C(\varepsilon)\,(1 + |x|)^{-1/2} \qquad \text{for all } x \in \mathbb{R}^2.$$

Proof. By the bootstrap, $\|f\|_{C^1} \le C(\varepsilon)$. The standard stationary-phase estimate for extension from a C^∞ curve in $\mathbb{R}^2$ with C^1 density gives $|Ef(x)| = O(|x|^{-1/2})$ for $|x| \ge 1$, and $|Ef(x)| \le (2\pi)^{1/2}\|f\|_{L^2} \le (2\pi)^{1/2}$ everywhere. □

9.2 Profile decomposition for the circle extension

The extension $E : L^2(S^1) \to L^6(\mathbb{R}^2)$ is not compact (Proposition 7.1), and the sole obstruction is the translation symmetry. The following decomposition, analogous to results of Merle–Vega and Shao for the paraboloid, captures this defect of compactness completely.

Theorem 9.3 (Profile decomposition)**.** ***[Conditional]*** *Let $f_j \in L^2(S^1)$ with $\|f_j\|_2 \le C$. After passing to a subsequence, there exist profiles $\phi^k \in L^2(S^1)$ $(k = 1, 2, \ldots)$ and group elements $g_j^k = (e^{i\alpha_j^k}, R_j^k, x_j^k) \in G$ satisfying the* divergence condition *$|x_j^k - x_j^l| \to \infty$ for all $k \ne l$, such that for each $K \ge 1$,*

$$f_j = \sum_{k=1}^{K} (g_j^k)^{-1} \cdot \phi^k + r_j^K, \tag{13}$$

with the decoupling identities

$$\|f_j\|_2^2 = \textstyle\sum_{k=1}^{K} \|\phi^k\|_2^2 + \|r_j^K\|_2^2 + o(1), \tag{14}$$

$$\|Ef_j\|_6^6 = \textstyle\sum_{k=1}^{K} \|E\phi^k\|_6^6 + \|Er_j^K\|_6^6 + o(1), \tag{15}$$

and $\|Er_j^K\|_{L^6} \to 0$ as $K \to \infty$ (after further subsequence extraction at each stage).

The L^2 decoupling (14) is elementary: for $k \ne l$, $\left\langle e^{-ix_j^k\cdot\omega}\phi^k,\, e^{-ix_j^l\cdot\omega}\phi^l \right\rangle = \int_{S^1} \phi^k\overline{\phi^l}\, e^{i(x_j^l - x_j^k)\cdot\omega}\, d\sigma \to 0$ by the Riemann–Lebesgue lemma on S^1.

The L^6 decoupling (15) is deeper. Since $E(e^{-ix_j^k\cdot\omega}\phi^k)(x) = (E\phi^k)(x - x_j^k)$, distinct profiles produce extensions centred at diverging locations. A Brézis–Lieb argument using the Herglotz wave decay $|E\phi^k(x)| = O(|x|^{-1/2})$ (valid once ϕ^k is smooth; for L^2 profiles, one approximates and uses density) then gives the L^6 asymptotic additivity.

The *construction* of the decomposition proceeds iteratively. The key input is an inverse Strichartz inequality for S^1:

Lemma 9.4 (Inverse Strichartz for S^1)**.** *If $\|f_j\|_2 = 1$ and $\|Ef_j\|_{L^6} \ge \delta > 0$, then there exist $x_j \in \mathbb{R}^2$ such that the re-centred sequence $\tilde{f}_j(\omega) := e^{ix_j\cdot\omega} f_j(\omega)$ has a subsequence converging weakly to some $\phi \ne 0$ in $L^2(S^1)$.*

Proof. Since $\|Ef_j\|_6^6 \ge \delta^6$, a covering argument gives $x_j \in \mathbb{R}^2$ with $\int_{B(x_j,1)} |Ef_j|^6 \ge c\delta^6$. Set $\tilde{u}_j(x) := Ef_j(x + x_j) = E\tilde{f}_j(x)$. Then $\int_{B(0,1)} |\tilde{u}_j|^6 \ge c\delta^6$.

Every Herglotz wave satisfies $(\Delta + 1)u = 0$ and $\|u\|_{L^\infty} \le (2\pi)^{1/2}\|f\|_2 = (2\pi)^{1/2}$. By interior Schauder estimates for $\Delta u + u = 0$, the family $\{\tilde{u}_j\}$ is equicontinuous

on $\overline{B(0,1)}$. Arzelà–Ascoli extracts a uniformly convergent subsequence $\tilde{u}_j \to u_\infty$ on $\overline{B(0,1)}$.

Meanwhile, $\tilde{f}_j \rightharpoonup \phi$ (Banach–Alaoglu, after subsequence). Weak convergence gives $E\tilde{f}_j(x) \to E\phi(x)$ pointwise for every x (since $\omega \mapsto e^{ix\cdot\omega} \in L^2(S^1)$). So $u_\infty = E\phi$. From $\int_{B(0,1)} |u_\infty|^6 \geq c\delta^6 > 0$, we get $\phi \neq 0$. □

Remark 9.5 (Simplification from compact frequency support). For the paraboloid $\{(\xi, |\xi|^2)\}$, profile decompositions must account for parabolic rescalings (a second family of symmetries). For S^1, the frequency set is compact with no scaling symmetry, so translations are the sole defect of compactness. Moreover, the inverse Strichartz lemma follows from elliptic regularity for $\Delta u + u = 0$, rather than from bilinear restriction estimates.

9.3 Profiles of critical sequences inherit the equation

The following result is the key structural input: if the original sequence consists of critical points, then each profile in the decomposition is itself a critical point.

Proposition 9.6 (Each profile is a critical point)**.** ***[Proved]*** *Let (f_j, Λ_j) be normalised critical points with $\Lambda_j \to \Lambda_\infty > 0$, and let $\phi^1, \phi^2, \ldots$ be the profiles from Theorem 9.3 (re-indexed so that $x_j^1 = 0$). Then each profile satisfies*

$$E^*(|E\phi^k|^4 E\phi^k) = \Lambda_\infty \phi^k. \tag{16}$$

In particular, $\psi^k := \phi^k / \left\|\phi^k\right\|_2$ is a normalised critical point with eigenvalue $\widetilde{\Lambda}_k = \Lambda_\infty / \left\|\phi^k\right\|_2^4$.

Proof. We prove (16) for $k = 1$; the case $k \geq 2$ follows by applying the translation $y_j^k := x_j^k - x_j^1$ (which diverges) and repeating the argument.

Set $v^l := E\phi^l$ and $w_j := Er_j$. Write $\tilde{f}_j := e^{ix_j^1\cdot\omega} f_j$ for the re-centred sequence, so $\tilde{u}_j := E\tilde{f}_j = v^1 + \sum_{l\geq 2} v^l(\cdot - y_j^l) + w_j$. The Euler–Lagrange equation, tested against arbitrary $h \in L^2(S^1)$, reads

$$\Lambda_j \left\langle \tilde{f}_j,\, h \right\rangle_{L^2(S^1)} = \int_{\mathbb{R}^2} |\tilde{u}_j|^4 \tilde{u}_j \, \overline{Eh} \, dx. \tag{17}$$

Left side. Since $\tilde{f}_j = \phi^1 + \sum_{l\geq 2} e^{-iy_j^l\cdot\omega}\phi^l + r_j$ and $\left\langle e^{-iy_j^l\cdot\omega}\phi^l,\, h \right\rangle \to 0$ by Riemann–Lebesgue ($|y_j^l| \to \infty$), the left side converges to $\Lambda_\infty \langle \phi^1,\, h\rangle$.

Right side. Decompose $|\tilde{u}_j|^4\tilde{u}_j = |v^1|^4 v^1 + N_j$, where N_j collects all terms containing at least one factor of $\eta_j := \tilde{u}_j - v^1 = \sum_{l\geq 2} v^l(\cdot - y_j^l) + w_j$. We show $\int N_j \, \overline{Eh} \to 0$.

Step (i): η_j vanishes locally in L^6. For each fixed $R > 0$: $\left\|v^l(\cdot - y_j^l)\right\|_{L^6(B(0,R))} = \left\|v^l\right\|_{L^6(B(-y_j^l,R))} \to 0$ as $j \to \infty$ (since $v^l \in L^6(\mathbb{R}^2)$ and $B(-y_j^l, R)$ escapes to infinity), and $\|w_j\|_{L^6} \to 0$. Hence $\|\eta_j\|_{L^6(B(0,R))} \to 0$.

Step (ii): individual translated-profile terms vanish. Fix $l \geq 2$. The typical cross term to estimate is $I_j^l := \int |v^1(x)|^4 \, v^l(x - y_j^l) \, \overline{Eh(x)} \, dx$. Split $\mathbb{R}^2$ into $B(0,R)$ and its complement.

- On $B(0,R)$: $|v^1| \leq (2\pi)^{1/2}$, so $|I_j^l|_{B(0,R)}| \leq C \left\| v^l(\cdot - y_j^l) \right\|_{L^6(B(0,R))} \|Eh\|_{L^6} \to 0$ by step (i).
- On $\mathbb{R}^2 \setminus B(0,R)$: $|I_j^l|_{\mathbb{R}^2 \setminus B}| \leq C \left\| v^l \right\|_6 \|Eh\|_{L^6(|x|>R)} \to 0$ as $R \to \infty$ (since $Eh \in L^6$ has vanishing tails).

An $\varepsilon/3$ argument in R and j gives $I_j^l \to 0$.

Step (iii): the remainder. Terms involving $w_j = Er_j$ are bounded by $C \|w_j\|_6 \|Eh\|_6 \to 0$ (using $\|v^1\|_\infty \leq C$ and Hölder).

Step (iv): higher-order cross terms. Every monomial in N_j has at least one factor η_j or $\bar{\eta}_j$. The same local-vanishing argument applies: on $B(0,R)$, the factor of η_j is small in L^6; on the complement, Eh is small in L^6.

Combining: the right side of (17) converges to $\int |v^1|^4 v^1 \, \overline{Eh} = \langle E^*(|E\phi^1|^4 E\phi^1), \, h \rangle$. Since h is arbitrary, $E^*(|E\phi^1|^4 E\phi^1) = \Lambda_\infty \phi^1$.

The rescaling: if $\alpha_k := \left\| \phi^k \right\|_2 > 0$, then $\psi^k := \phi^k / \alpha_k$ satisfies $E^*(|E\psi^k|^4 E\psi^k) = \alpha_k^{-5} E^*(|E\phi^k|^4 E\phi^k) = \Lambda_\infty \alpha_k^{-5} \phi^k = (\Lambda_\infty / \alpha_k^4) \psi^k$. □

9.4 Existence of a maximiser

Theorem 9.7 (Existence of extremisers for S^1 restriction)**.** ***[Proved]*** *The sharp restriction constant $\mathcal{R}$ is attained: there exists $f_* \in L^2(S^1)$ with $\|f_*\|_2 = 1$ and $\|Ef_*\|_{L^6} = \mathcal{R}$.*

Proof. Let f_j be a maximising sequence: $\|f_j\|_2 = 1$, $\Lambda_j := \|Ef_j\|_6^6 \to \mathcal{R}^6 > 0$. Apply the profile decomposition (Theorem 9.3).

Step 1: at least one profile. Since $\left\| Er_j^K \right\|_6 \to 0$ as $K \to \infty$ while $\Lambda_j \to \mathcal{R}^6 > 0$, the decoupling (15) forces $\phi^1 \neq 0$.

Step 2: no splitting. By the restriction inequality, $\left\| E\phi^k \right\|_6^6 \leq \mathcal{R}^6 \left\| \phi^k \right\|_2^6$. Set $\alpha_k := \left\| \phi^k \right\|_2$ and $a_k := \alpha_k^2$. The decoupling gives $\mathcal{R}^6 \leq \mathcal{R}^6 \sum_k a_k^3$. From (14), $\sum_k a_k \leq 1$.

If $M \geq 2$ profiles are nonzero, all $a_k > 0$ and $\sum a_k \leq 1$. But the multinomial expansion $\left(\sum_k a_k\right)^3 = \sum_k a_k^3 + 3 \sum_{k \neq l} a_k^2 a_l + 6 \sum_{k<l<m} a_k a_l a_m$ shows $\sum_k a_k^3 < \left(\sum_k a_k\right)^3 \leq 1$ (since the cross terms are strictly positive). This gives $\mathcal{R}^6 < \mathcal{R}^6$, a contradiction.

Step 3: convergence. With a single profile: $f_j = g_j^{-1} \cdot \phi^1 + r_j$, $\|r_j\|_2 \to 0$, $\|\phi^1\|_2 = 1$. Hence $g_j \cdot f_j \to \phi^1$ strongly, and ϕ^1 is a maximiser. □

Remark 9.8 (The homogeneity gap)*.* The argument uses $6 > 2$: the L^6 norm is superquadratic in f, making $\sum a_k^3 < (\sum a_k)^3$ when there are multiple profiles. This mechanism is specific to maximising sequences and does not extend to sub-maximal critical points.

9.5 The splitting obstruction for general critical sequences

For a general sequence of critical points with $\Lambda_j \to \Lambda_\infty \in [\varepsilon, \mathcal{R}^6)$, the strict-convexity argument breaks down: multi-profile decompositions are *algebraically consistent* with the Euler–Lagrange equation.

Proposition 9.9 (Structure of multi-profile decompositions at critical sequences)**.** ***[Proved]*** *Suppose a sequence of normalised critical points with $\Lambda_j \to \Lambda_\infty > 0$ admits a profile decomposition with $M \geq 2$ nonzero profiles $\phi^1, \ldots, \phi^M$. Then:*

1. *Each normalised profile $\psi^k = \phi^k/\alpha_k$ (where $\alpha_k := \left\|\phi^k\right\|_2$) is a critical point with eigenvalue $\widetilde{\Lambda}_k = \Lambda_\infty/\alpha_k^4 > \Lambda_\infty$.*
2. *The number of profiles satisfies $M \leq (\mathcal{R}^6/\varepsilon)^{1/2}$.*
3. *The identity $\sum_k \alpha_k^6 \widetilde{\Lambda}_k = \Lambda_\infty \sum_k \alpha_k^2 = \Lambda_\infty$ holds automatically, so the energy decomposition produces no contradiction.*

Proof. (1) Proposition 9.6 gives $E^*(|E\phi^k|^4 E\phi^k) = \Lambda_\infty \phi^k$. Rescaling: $E^*(|E\psi^k|^4 E\psi^k) = (\Lambda_\infty/\alpha_k^4)\psi^k$, so $\widetilde{\Lambda}_k = \Lambda_\infty/\alpha_k^4$. Since $\alpha_k^2 < \sum_l \alpha_l^2 = 1$, we have $\alpha_k < 1$ and $\widetilde{\Lambda}_k > \Lambda_\infty$.

(2) From $\widetilde{\Lambda}_k \leq \mathcal{R}^6$: $\alpha_k^4 \geq \Lambda_\infty/\mathcal{R}^6 \geq \varepsilon/\mathcal{R}^6$, so $\alpha_k^2 \geq (\varepsilon/\mathcal{R}^6)^{1/2}$. The constraint $\sum_k \alpha_k^2 = 1$ gives the bound on M.

(3) $\sum_k \alpha_k^6 \widetilde{\Lambda}_k = \sum_k \alpha_k^6 \cdot \Lambda_\infty/\alpha_k^4 = \Lambda_\infty \sum_k \alpha_k^2 = \Lambda_\infty$. □

Part (3) is the central difficulty: **the Euler–Lagrange equation alone does not prevent a critical sequence from splitting into bubbles at diverging locations, each individually a critical point at a higher eigenvalue level.** This is the open content of Theorem 9.10:

Theorem 9.10 (Compactness modulo G)**.** ***[Open]*** *Let (f_j, Λ_j) be normalised critical points with $\Lambda_j \geq \varepsilon > 0$. Then there exist $g_j \in G$ and a subsequence such that $g_j \cdot f_j \to f_\infty$ strongly in $L^2(S^1)$ and $\Lambda_j \to \Lambda_\infty = \|Ef_\infty\|_6^6$.*

Equivalently: every critical sequence has **exactly one nonzero profile** in its decomposition.

9.6 Approaches to rule out splitting

We describe three strategies that may close the gap.

(A) Descending induction on critical values. At the maximal level $\Lambda = \mathcal{R}^6$, Theorem 9.7 proves single-profile concentration. Suppose inductively that compactness modulo G holds for critical sequences with $\Lambda_j \to \Lambda'$ for all $\Lambda' > \Lambda_\infty$. A splitting at level Λ_∞ produces profiles at levels $\widetilde{\Lambda}_k > \Lambda_\infty$, where the inductive hypothesis already gives finiteness of shapes. The splitting identity $\sum_k \alpha_k^2 = 1$, $\widetilde{\Lambda}_k = \Lambda_\infty/\alpha_k^4$ imposes

algebraic constraints relating the profile masses to the (now finitely many) critical values above Λ_∞.

Difficulty: the base case gives one shape (the maximiser); but the inductive step requires both finiteness and nondegeneracy at every intermediate level, creating a dependency between this theorem and the results of Section 8. Breaking this circularity requires either a direct nondegeneracy argument at all levels, or an independent route to single-profile concentration.

(B) Pohozaev–Morawetz monotonicity. A localised identity for $u = c(|u|^4 u * \widehat{\sigma})$ could yield a monotonicity formula controlling the spatial distribution of $|u|^6$ and preventing mass separation. The starting point would be to multiply the equation by a radial weight $a(|x|)\,\bar{u}$ and integrate; the resulting identity relates $\int a'|u|^6$ to boundary terms and the nonlocal convolution structure.

Difficulty: the convolution against $\widehat{\sigma}$ (which decays only like $|x|^{-1/2}$) introduces long-range interactions that resist clean localisation.

(C) Unique continuation for the integrated equation. If one can show that two critical points ϕ^1, ϕ^2 at levels $\widetilde{\Lambda}_1, \widetilde{\Lambda}_2 > \Lambda_\infty$ cannot coexist as profiles of a common critical sequence—i.e., the relation

$$f_j \approx e^{-ix_j^1\cdot\omega}\phi^1 + e^{-ix_j^2\cdot\omega}\phi^2, \qquad |x_j^1 - x_j^2| \to \infty,$$

contradicts $E^*(|Ef_j|^4 Ef_j) = \Lambda_j f_j$ at order beyond the profile-level Euler–Lagrange—this would rule out splitting. The relevant obstruction would arise from cross terms at intermediate spatial scales $|x| \sim |x_j^1 - x_j^2|^{1/2}$ where the two Herglotz waves overlap at polynomial (not exponential) level.

Remark 9.11 (Sufficiency for identifying $\mathcal{R}$). For the purpose of *computing* the sharp constant $\mathcal{R}$, the full Theorem 9.10 is not needed. Theorem 9.7 gives existence of a maximiser, and if the maximiser is unique modulo G (e.g., the constant function $f_0 = (2\pi)^{-1/2}$), then $\mathcal{R} = \|Ef_0\|_{L^6}$ and the problem is solved. The open compactness theorem would additionally yield finiteness of *all* critical values above ε, which is a stronger structural result.

Remark 9.12 (Accumulation of critical values at zero). Whether critical values can accumulate at 0 is unknown. The compactness + nondegeneracy programme gives finiteness only above a fixed threshold. For the sharp constant, it suffices to work above any ε smaller than the maximum critical value.

10 From the Three Inputs to the Sharp Constant

Theorem 10.1 (Conditional finiteness of shapes). ***[Conditional]*** *Assume:*

1. *Compactness modulo G (Theorem 9.10) holds for critical points with $\Lambda \geq \varepsilon$;*

2. *Every such critical point is nondegenerate modulo G (i.e.,* $\ker L = T_{f_0}(G \cdot f_0)$ *on the tangent space to the unit sphere).*

Then the set of shapes $\{[f, \Lambda] \in \mathcal{Z} : \Lambda \geq \varepsilon\}$ *is finite. In particular, the set of critical values above* ε *is finite.*

Proof. By compactness modulo G, the quotient $\mathcal{Z}_\varepsilon := \{f : \text{critical}, \Lambda \geq \varepsilon\}/G$ is compact.

By nondegeneracy: at each $[f_0] \in \mathcal{Z}_\varepsilon$, choose a slice S_{f_0} complementary to $T_{f_0}(G \cdot f_0)$ in the tangent space to $\|f\|_2 = 1$. The condition $\ker L|_{S_{f_0}} = \{0\}$ plus Fredholm index zero implies $L|_{S_{f_0}}$ is invertible with bounded inverse $\left\| L^{-1}|_{S_{f_0}} \right\| \leq \mu_{f_0}^{-1}$. By the implicit function theorem, $[f_0]$ is an isolated point of $\mathcal{Z}_\varepsilon$.

A compact set all of whose points are isolated is finite. □

Corollary 10.2 (Identification of the sharp constant)**.** *Under the hypotheses of Theorem 10.1, the sharp restriction constant is*

$$\mathcal{R} = \max\{\Lambda_j^{1/6} : \Lambda_j \geq \varepsilon\}$$

for any ε *smaller than all critical values in the finite list. The maximiser is the shape achieving the largest* Λ_j*.*

10.1 What remains for the sharp constant on S^1

Two paths are available, depending on the desired strength of the result.

Path 1: Identify $\mathcal{R}$ (the sharp constant).

1. **Verify Bessel integrals.** Check $\mathcal{J}_n \neq \mathcal{J}_0$ and $5\mathcal{J}_n \neq \mathcal{J}_0$ for $n = 2, 3, \ldots, n_*$ by interval arithmetic. This establishes nondegeneracy at the constant solution. *Routine numerical computation.*

2. **Show the constant is the unique maximiser.** Theorem 9.7 gives existence of a maximiser. If one can prove that the constant function $f_0 = (2\pi)^{-1/2}$ is the unique maximiser modulo G—e.g., by a rearrangement or symmetrisation argument, or by ruling out non-radial maximisers via the eigenvalue analysis—then $\mathcal{R} = \|Ef_0\|_{L^6}$ and the programme is complete. *This bypasses compactness of the full critical set.*

Path 2: Finiteness of all critical values above ε (the full structural result).

1. Bessel integrals (same as above).

2. **Rule out multi-profile splitting (Theorem 9.10).** The profile decomposition is in place (Theorem 9.3), and each profile inherits the Euler–Lagrange equation (Proposition 9.6). The open step is to show that no critical sequence admits more than one profile. Section 9 identifies three strategies: descending induction, Pohozaev–Morawetz monotonicity, and unique continuation for cross terms. *Requires new analytic input.*

3. **Enumerate non-constant shapes (if any).** Either prove that the constant is the only critical shape, or verify nondegeneracy at all other shapes.

Part III
Extensions and Supplementary Material

11 The Higher-Dimensional Transversality Route

For $n \geq 2$ and the equation $\rho = cT(\rho)$ on $L^2(S^{n-1})$, Proposition 4.1 forces $\widehat{u}$ onto S^{n-1}, and the problem reduces to the incidence-fibre equation (3). The symmetry group is $G = \mathbb{R}^n \rtimes SO(n)$, acting on densities by $(x_0, R) \cdot \rho(\omega) = e^{-ix_0\cdot\omega}\rho(R^{-1}\omega)$.

11.1 The five-step programme

The route to finiteness of shapes requires five analytic inputs. We state each and assess its status for general n.

Step 1: Smoothing / Fredholm estimate. [Open] The target is a multilinear smoothing estimate

$$A_k : H^s(S^{n-1})^k \to H^{s+\varepsilon}(S^{n-1}), \qquad \|A_k(\rho_1, \ldots, \rho_k)\|_{H^{s+\varepsilon}} \leq C_s \prod_{j=1}^{k} \|\rho_j\|_{H^s} ,$$

for some $\varepsilon > 0$, where A_k is the k-linear operator behind the incidence-fibre equation. A weaker but sufficient target is compactness of A_k on bounded sets.

For the S^1 problem, the truncation argument (Theorem 7.3) bypasses this step entirely. For $n \geq 2$, the proof must exploit the curvature of S^{n-1} and the transversality of the constraint $\omega_1 + \cdots + \omega_k = \omega$. The key obstruction is the existence of *degenerate fibres*; see Section 11.2.

Step 2: Regularity bootstrap. **[Open]** Show that weak L^2 solutions are automatically smooth: $\rho \in L^2(S^{n-1}),\ \rho = cT(\rho) \implies \rho \in H^s$ for all s. If Step 1 gives $T : L^2 \to H^\varepsilon$, iteration yields the bootstrap.

Step 3: Local slices for the symmetry action. **[Open]** At a solution ρ, construct a complement S_ρ to the orbit tangent

$$\mathfrak{g} \cdot \rho = \{-i(a \cdot \omega)\rho - (A\omega) \cdot \nabla_{S^{n-1}}\rho : a \in \mathbb{R}^n,\ A \in \mathfrak{so}(n)\}.$$

This must be done separately on orbit-type strata (radial, zonal, fully asymmetric solutions have different stabiliser dimensions).

Step 4: Nondegeneracy modulo symmetry. **[Open]** Verify $\ker L_\rho = \mathfrak{g} \cdot \rho$ at each solution, where $L_\rho = I - ckA_k(\rho, \ldots, \rho, \cdot)$. For the S^1 problem at the constant solution, this reduces to Bessel integrals (Theorem 8.4).

Step 5: Compactness modulo G. **[Open]** Prove sequential compactness of the solution set modulo G:

$$\rho_j = cT(\rho_j) \implies \exists g_j \in G :\ g_j \cdot \rho_j \to \rho_\infty.$$

This requires a priori bounds ($\|\rho\| \le R$), a non-collapse lower bound ($\|\rho\| \ge r > 0$), and a no-escape condition for translations. Rotations are compact; translations are the genuine noncompact obstruction.

11.2 Degenerate fibres

The fibre $V_k(\omega) = \{(\omega_1, \ldots, \omega_k) \in (S^{n-1})^k : \sum \omega_j = \omega\}$ is generically a smooth manifold of dimension $(n-1)k - (n-1+1) = (n-1)(k-1) - 1$... when the differential of $\Sigma : \omega_1 + \cdots + \omega_k$ has full rank. Degeneracies occur when all tangent lines $T_{\omega_j} S^{n-1}$ are parallel.

Example for $k = 5$ on S^1: The configuration $(\omega_0, \omega_0, \omega_0, -\omega_0, -\omega_0)$ with output $\omega = \omega_0$ is degenerate. Here $d\Sigma = 3(\text{tangent at } \omega_0) - 2(\text{tangent at } -\omega_0)$ has rank 1 instead of 2 (the tangents at ω_0 and $-\omega_0$ are parallel on S^1). This is a codimension-1 degeneracy, producing an integrable singularity $\sim |t|^{-1/2}$ in the fibre density.

In higher dimensions, the degenerate locus can have higher codimension (tangents at ω_0 and $-\omega_0$ are parallel in $\mathbb{R}^n$ only when $n = 1$; for $n \ge 2$, anti-parallel tangent *spaces* are generically transverse). This suggests that the smoothing estimate may be easier for $n \ge 3$ than for $n = 2$.

11.3 Orbit dimension and radial rigidity

Proposition 11.1 (Orbit dimension, $n = 3$)**.** *For $G = \mathbb{R}^3 \rtimes SO(3)$ ($\dim G = 6$), the orbit dimension of a solution u is $6 - \dim \mathrm{Stab}_G(u)$. Specifically:*

- *Radial: $d = 3$ (only translations).*
- *Zonal (one axis of symmetry): $d = 5$.*
- *Fully asymmetric: $d = 6$.*

Conjecture 11.2 (Radial rigidity)**.** *For the equation $u = c(|u|^{2(k-1)} u * \widehat{\sigma})$ on $\mathbb{R}^3$, every non-trivial solution is rotationally invariant about some point (its "centre of mass"). If true, all solutions belong to 3-dimensional translation orbits, and the shape count reduces to counting radial profiles.*

Warning 11.3. Conjecture 11.2 is currently *unproved*. The preceding sections on the $n = 3$ problem treated it as established; it should be regarded as a useful heuristic until proved.

12 Supplementary: Gegenbauer Expansion and Galerkin Models

12.1 Gegenbauer expansion

[Heuristic]

For completeness, we record the formal harmonic decomposition. Let $\rho(\omega) = \sum_\ell a_\ell C_\ell^\lambda(\omega \cdot \eta)$ with $\lambda = (n-2)/2$. The product of Gegenbauer polynomials linearises as $C_{\ell_1}^\lambda C_{\ell_2}^\lambda = \sum_j \gamma_j(\ell_1, \ell_2) C_j^\lambda$. If T were a spherical convolution, the Funk–Hecke theorem would give $C_\ell^\lambda *_S \sigma = \lambda_\ell(\sigma) C_\ell^\lambda$. Substituting into $\rho = cT(\rho)$ and equating coefficients would give an algebraic system for $\{a_\ell\}$.

However, as noted in Warning 4.3, the incidence-fibre operator is *not* spherical convolution. The Gegenbauer expansion gives a useful *model* but does not rigorously reduce to finitely many coefficients without separate justification.

12.2 Galerkin approximation

Let $H_L = \mathrm{span}\{Y_{\ell m} : 0 \le \ell \le L\}$ and P_L the $L^2(S^{n-1})$ projection. The projected equation $P_L F(\rho_L) = 0$ is a polynomial system of degree k in $N(L) = (L+1)^2$ unknowns (for $n = 3$).

Proposition 12.1 (Galerkin shape count)**.** *After imposing algebraic slice conditions, the number of Galerkin shapes at truncation L is bounded by the BKK mixed volume of the sliced Newton polytopes, in particular by $k^{N(L)}$ (Bézout).*

Warning 12.2 (Limit caveat). This is a statement about the fixed-L model. It does not imply finiteness of true shapes as $L \to \infty$ without uniform a priori bounds, no-escape compactness, and stability of roots under high-mode perturbation.

Remark 12.3 (How Galerkin helps). A Galerkin computation becomes rigorous if paired with a high-mode bound $\|(I - P_L)T(\rho)\|_X \le \varepsilon_L \Phi(\|\rho\|_X)$ with $\varepsilon_L \to 0$, plus an inverse bound on the sliced Jacobian at each computed root. A Newton–Kantorovich argument would then validate nearby true solutions and certify completeness.

13 Supplementary: Validated Search Strategy

Assuming the transversality route is completed (modulo G the solution set is finite), the next goal is to *find every shape* by a finite, checkable search.

13.1 Constants needed

1. **A priori radius:** $\|\rho\|_X \le R(c, k)$.
2. **Non-collapse:** $\|\rho\|_X \ge r(c, k) > 0$.
3. **Regularity/mesh modulus:** $\|\rho\|_{C^m} \le B_m$.
4. **Validated quadrature:** error $q_h(B_m) \to 0$ for the fibre integral on a mesh of size h.
5. **Slice conditioning:** $\|L_\rho \eta\|_X \ge \mu(\rho) \|\eta\|_X$ on S_ρ, with $\mu_* := \inf_\rho \mu(\rho) > 0$.
6. **Lipschitz:** $\|DF(\rho) - DF(\tilde{\rho})\| \le M \|\rho - \tilde{\rho}\|_X$.

13.2 Separation and packing

From the conditioning bound, the quantitative isolation scale is $\delta_* \simeq \mu_*/M$: two distinct shapes cannot lie within δ_* of each other in the slice. A crude packing bound is

$$\#\{\text{shapes}\} \le \left(1 + \frac{2R_h}{\delta_* - 2\eta_h - 2q_h}\right)^{N_{\text{mesh}}(h)},$$

provided $\delta_* > 2\eta_h + 2q_h$.

13.3 The search pipeline

1. Choose a quasi-uniform mesh on S^{n-1} (or directly on S^1 for the circle problem) and fix gauge conditions.

2. Represent candidates by interval boxes for nodal values plus global regularity bounds.
3. For each box, enclose the residual $F(\rho)$ at mesh points using validated quadrature.
4. Discard boxes with nonzero residual enclosure.
5. Apply Krawczyk/interval Newton to surviving boxes.
6. Validate uniqueness and lift to the full space.
7. Certify completeness: every box discarded or assigned.

13.4 Main difficulties

1. **Translation gauge:** $e^{-ix\cdot\omega}\rho(\omega)$ creates oscillation as $|x| \to \infty$. A canonical centre must be fixed.
2. **Uniform conditioning:** $\mu_* > 0$ may be the hardest constant to obtain without knowing all solutions.
3. **Mesh consistency:** nodal errors must not hide solutions between mesh points.
4. **Dimension:** the effective search dimension grows with mesh size. Symmetry decomposition and adaptive refinement are essential.

13.5 Specialisation to S^1

For the circle problem, the validated search simplifies considerably:

- The mesh is a grid on $[0, 2\pi)$ with N equally spaced points.
- The gauge fixes phase, rotation of S^1, and one translation.
- The fibre integral reduces to a trigonometric integral.
- The Jacobian involves the linearised operator L, which decomposes into Fourier modes for radial u_0.

The direct certified programme for the sharp constant is:

1. Prove a priori bounds: $\|f\|_{L^2} = 1$, $\|f\|_{C^m} \le B_m$, $0 < \Lambda_{\min} \le \Lambda \le C_{\mathrm{ST}}^6$.
2. Fix gauges (phase, rotation, translation).
3. Mesh S^1; represent f by interval boxes plus derivative bounds.
4. For each (f, Λ) box, enclose the residual $E^*(|Ef|^4 Ef)(\omega_i) - \Lambda f(\omega_i)$.
5. Discard or validate via interval Newton.
6. Sort validated Λ_j intervals; certify the ordering.

14 Summary: The Logical Dependency Graph

We summarise the dependencies and status of the full programme.

Proved:

- Spectral support forcing ($\widehat{u} \subseteq S^1$)
- 1D case complete
- Compactness of linearised operator K_{u_0} on $L^2(S^1)$
- Fredholm index zero for $L = K - \Lambda I$
- Eigenvalue decomposition at the constant solution
- Nondegeneracy $\Leftrightarrow$ Bessel integrals (11)
- $\mathcal{J}_n \to 0$ (finitely many integrals to check)
- Regularity bootstrap for critical points
- Inverse Strichartz inequality for S^1 extension
- Profiles of critical sequences are critical points
- Existence of a maximiser (Theorem 9.7)

Next concrete steps:

- Compute $\mathcal{J}_n$ for $n = 2, 3, \ldots, n_*$ by interval arithmetic; verify (11)
- Prove the constant is the unique maximiser modulo G (this alone gives $\mathcal{R}$)

Open (harder):

- Rule out multi-profile splitting for non-maximising critical sequences (Theorem 9.10)
- Enumeration / uniqueness of non-constant shapes

Conditional conclusion:

- Compactness + nondegeneracy $\Longrightarrow$ finitely many critical values Λ_j
- $\mathcal{R} = \max_j \Lambda_j^{1/6}$

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*Notes compiled from working discussions. Statements marked **[Proved]** carry complete proofs; statements marked **[Conditional]** depend on explicitly stated hypotheses; statements marked **[Open]** require new analytic input.*